Hints: Linear algebra – 3. Eigenvalues, diagonalisability and the Jordan form

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Diagonalisability. If A is diagonalisable with distinct eigenvalues $\lambda_1, \dots, \lambda_s$, then $\prod(A - \lambda_i E) = O$ (check it on the eigenbasis). Conversely, if $\mu_A = \prod(x - \lambda_i)$ with the λ_i distinct, use Bezout (or Lagrange interpolation polynomials p_i with $\sum p_i = 1$) to get $V = \bigoplus \ker(A - \lambda_i E)$.

Hint 2. Frobenius form. Regard V as a $K[x]$-module (x acts as A). Equivalently, reduce the polynomial matrix $xE - A$ by elementary row and column operations over $K[x]$ to the Smith form $\operatorname{diag}(1, \dots, 1, f_1, \dots, f_k)$ with $f_1 \mid \cdots \mid f_k$. The module splits into cyclic pieces $K[x]/(f_i)$, and on each piece A acts by the companion matrix of f_i.

Hint 3. So $A \sim B$ over K if and only if $xE - A$ and $xE - B$ have the same invariant factors. These are determined by the gcds d_k of the $k \times k$ minors (f's are ratios d_k/d_{k-1}). Gcds of polynomials with coefficients in K do not change when you pass to a larger field (Euclid's algorithm runs inside $K[x]$). Hence similarity over an extension implies similarity over K.

Problem 2

Hint 1. Let C be the $n \times n$ tridiagonal matrix with zeros on the diagonal and ones just above and below it (the adjacency matrix of a path). Compute C^2.

Hint 2. $C^2 = A$. The eigenvalues of C are $2\cos\frac{k\pi}{n+1}$, $k = 1, \dots, n$: try eigenvectors $(\sin\frac{jk\pi}{n+1})_j$.

Hint 3. So the eigenvalues of A are $4\cos^2\frac{k\pi}{n+1}$, $k = 1, \dots, n$. The values for k and $n + 1 - k$ coincide. Hence for even n each of the $n/2$ distinct values has multiplicity 2; for odd n the value 0 (at $k = \frac{n+1}{2}$) is simple and the remaining $\frac{n-1}{2}$ values are double.

Problem 3

Hint 1. μ_A divides $f(x) = x^{2n} + x^n + 2n$. By Fact 1 / Problem 1 it suffices to show that f has no multiple roots.

Hint 2. $f'(x) = nx^{n-1}(2x^n + 1)$. Check that neither $x = 0$ nor $x^n = -\frac{1}{2}$ is a root of f.

Problem 4

Hint 1. A real eigenvalue of the real matrix AB has a real eigenvector: $ABv = \lambda v$, $v \neq 0$. Rewrite it as $Bv = \lambda A^{-1}v$ and take the inner product with v.

Hint 2. $\langle Bv, v\rangle > \|v\|^2$, while $0 < \langle A^{-1}v, v\rangle < \|v\|^2$, because the eigenvalues of A^{-1} lie in $(0, 1)$. So $\lambda = \langle Bv, v\rangle / \langle A^{-1}v, v\rangle > 1$.

Problem 5

Hint 1. By Theorem 3, $\sum_{i<j} \lambda_i\lambda_j$ is the sum of the principal 2×2 minors of A.

Hint 2. That sum equals $\sum_{i<j}(a_{ii}a_{jj} - a_{ij}^2)$, so the difference is $\sum_{i<j} a_{ij}^2 \geqslant 0$. Equality holds exactly for diagonal matrices.

Problem 6

Hint 1. $\langle Av, v\rangle$ depends only on the symmetric part $S = \frac{1}{2}(A + A^T)$. Show that $S = E$, so $A = E + K$ with K skew-symmetric.

Hint 2.

(a) The eigenvalues of a real skew-symmetric K are purely imaginary, and K is normal, hence diagonalisable over $\mathbb{C}$. If all eigenvalues of $E + K$ are real, all eigenvalues of K are 0, so $K = O$. (b) Take K block-diagonal with 1012 blocks $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Problem 7

Hint 1. Yes, they exist. Try two square-zero matrices of rank 1 with the same image but different kernels, e.g. $A = E_{12}$ and $B = E_{13}$ (matrix units). Check that they commute.

Hint 2. In a common Jordan basis f_1, f_2, f_3, each of A, B maps exactly one basis vector onto the basis vector immediately preceding it and kills the other two. Since $\operatorname{im} A = \operatorname{im} B$, the image vector is the same basis vector for both.

Hint 3. Since $\ker A \neq \ker B$, the vectors sent onto it are different for A and B. Both would have to stand immediately after it in the basis, which is impossible.

Problem 8

Hint 1. You need a basis e_1, e_2 with $Me_1 = 0$, $Me_2 = e_1$, $Ne_1 = e_2$, $Ne_2 = 0$. Start with $e_1 \neq 0$ in $\ker M$ and set $e_2 = Ne_1$.

Hint 2. $Me_2 = MNe_1 = (E - NM)e_1 = e_1$, which also shows e_1, e_2 are independent, and $Ne_2 = N^2e_1 = 0$. Take A with columns e_1, e_2.